

ME 623: Fundamentals of Fracture Mechanics

Course Code: ME 623

Course Name: Fundamentals of Fracture Mechanics

L-T-P-C: 3 credits

Approval: 9th Senate (specialisation elective in the M.Tech Energy Engineering curriculum); one-time approval (OTA) under code ME 623 recommended by the 18th BoA and approved in the 14th Senate

Course Contents

- **Introduction:** Why structures fail?, An atomic view, Energy criterion, Stress intensity, Effect of material properties on fracture, Modes of failure.
- **Linear Elastic Fracture Mechanics:** An atomic view of fracture, Effect of flaws on stress concentration, Griffith theory of fracture, Energy release rate, Instability and the R curve, Stress analysis of cracks, Stress Intensity Factor (SIF), Determination SIF of different geometries, Crack tip plasticity, Irwin approach, Plane strain fracture, Mixed mode fracture.
- **Elastic-Plastic Fracture Mechanics:** Crack tip opening displacement (CTOD), J contour integral, Relationship between J and CTOD, Resistance curve, Cleavage fracture, failure criterion, Three-dimensional effect, Crack arrest.
- **Fracture Mechanisms in Metals and Nonmetals:** Ductile fracture, Void nucleation, Void growth, coalescence, Ductile crack growth, Brittle Fracture, Cleavage, Mechanisms of cleavage initiation, Transgranular and Intergranular fracture, Fracture Mechanisms in Nonmetals.
- **Fracture Toughness Testing:** Specimen configurations and orientations, K_{IC} testing, CTOD testing, Measurement of J-critical, Determination of critical G in Mode I and Mode II.
- **Fatigue Crack Propagation and Environmental Assisted Fracture Failure:** Fatigue crack growth, Crack closure, A short introduction to environmental assisted fracture failure.
- **Computational Fracture Mechanics:** Modeling and analysis.

Text Book

1. T.L. Anderson, *Fracture Mechanics – Fundamentals and Applications*, Taylor and Francis Group.
2. Prashant Kumar, *Elements of Fracture Mechanics*, Tata McGraw Hill, New Delhi, India.
3. G.E. Dieter, *Mechanical Metallurgy*, McGraw Hill.